TEXORA · SESSIONAL NOTES · STUDY SHEET

Case Study in Apparels Manufacturing

Working one complete order as a case — from order sheet to shipment audit — and drawing the engineering lessons from its numbers. The case is a teaching case built for this study; its figures are internally consistent, not a report of a real factory.

AM-4202 · ADVANCED APPAREL MANUFACTURING-II SESSIONAL · STUDY 6 · LEVEL 4 TERM 2

1. Aim and place of the study

Aim: to follow a single apparel order through every department, recording the decision and the calculation at each hand-over, so that the separate skills of this sessional — machines, production, CAD, balancing, costing — are seen working as one system.

2. The case: order at a glance

ITEM	CASE DATA
Product	Men's basic long-sleeve shirt, sizes S–XL in ratio 1:2:2:1, one colour — the shirt of Study 5.
Quantity	2,400 pieces (200 dozen) — deliberately the lot size of the TE-3107 AQL example.
Fabric	Cotton poplin, 150 cm width; consumption from the Study 3 method; fabric and trims received and inspected before cutting.
Delivery	Single shipment; final audit before packing release.

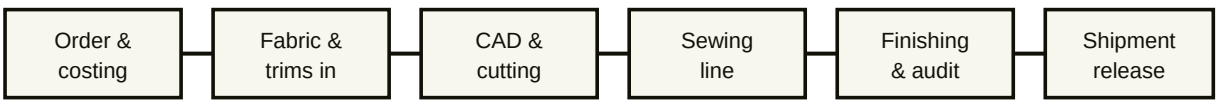


Figure 1 — The order's path through the factory. Each arrow is a hand-over with its own document and its own check; the case follows them in order.

3. Stage by stage

Order and costing. The order was costed by the Study 5 method: production cost Tk 459.00 per piece, FOB Tk 491.59 after the 2% wastage allowance and 5% margin. The merchandiser's first check is quantity against capacity: at the line's target of 800 pieces per day the order needs **3 line-days** of sewing (2,400 ÷ 800), plus cutting before and finishing/audit after — the delivery promise is built from these numbers, not from optimism.

Materials in. Fabric was shade-checked roll to roll and a 4-point inspection sample taken; trims were counted against the trim card. Two rolls fell outside the approved shade band and were segregated for a later, shade-tolerant order — recorded as a materials note, because a shade claim at final audit is unwinnable once garments are sewn.

CAD and cutting. Patterns were graded by CAD (Study 4) and the marker for the size ratio 1:2:2:1 planned at 1.48 m set width. Cutting followed the bundle system, each bundle ticketed with size and shade lot so that parts from one roll stayed together through sewing — the control that prevents shade variation *within* a garment.

Sewing line. The line ran exactly as balanced in Study 5: total SMV 18.60 min, pitch 0.60 min, 31 operators on operations plus helpers and inspection, 34 attended in all. Output held the target of 800 pieces per day and line efficiency measured **91.2%** — (18.60 × 800) ÷ (34 × 480) — over the three days. First-day output was lower (a normal learning curve on a style change), recovered on day two by moving one helper to the collar-attaching station, the day's tightest point: a small, observed re-balance rather than a new plan.

In-line quality. End-of-line checkers inspected 600 garments across the run and recorded 40 defects:

$$\text{DHU} = 40 \div 600 \times 100 = 6.67$$

above the factory's working target of 5 — a signal, not a shipment decision (the audit decides shipment). The defect pattern is analysed in Study 7; the line response in this case was a same-day check of needle condition and a re-briefing at the two operations producing most defects.

Finishing and final audit. After pressing and packing preparation, the final audit applied the ISO 2859-1 chain of TE-3107: lot 2,400 at General Inspection Level II → code letter K → sample **n = 125**; acceptance numbers at AQL 2.5 (majors) **Ac 7 / Re 8** and at AQL 4.0 (minors) **Ac 10 / Re 11**, critical defects Ac 0. The auditors found **5 major-defective** and **9 minor-defective** garments and no criticals: 5 ≤ 7 and 9 ≤ 10, so the lot **passed** both classes and was released for shipment.

4. What the case teaches

- **One set of numbers.** The same SMV (18.60) drives the target, the staffing, the efficiency and the CM inside the costing; the same lot size (2,400) drives cutting quantities and the audit plan. A case — like a factory — fails first at the joints where numbers are re-invented.
- **Checks sit where correction is cheap.** Shade was rejected at the fabric stage for the price of segregation; the same fault found at final audit would have cost the order.
- **Signals versus decisions.** DHU 6.67 was a signal that triggered diagnosis; the AQL audit was the decision that released the shipment. Confusing the two either ships bad lots or stops good ones.
- **Documentation is the memory.** Every stage closed with a record (costing sheet, inspection note, marker record, audit report) — Study 7 turns those records into the critical analysis.

Precautions

- Keep the case honest: where the case assumes a value (efficiency, defect counts), the assumption is stated — never present teaching-case figures as measurements of a real factory.
- Do not "improve" a case after the fact: if a stage's numbers embarrass the plan, that tension is the teaching content.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why was the order's delivery promise built from 800 pieces per day rather than from the machine speeds of Study 1?
2. The audit had found 8 major-defective garments instead of 5. State the decision and the next steps.
3. What would DHU have been if 45 defects were found in the same 600 inspections, and what is DHU used for?
4. Which single document connects the costing, the line plan and the audit in this case?

Answer key

1. Machine speed is a theoretical maximum for one operation; delivery depends on the whole balanced line's demonstrated output per day — SMV, pitch, staffing and efficiency together — which is what 800 pieces per day represents.
2. Majors 8 ≥ Re 8: the lot fails on majors (minors still pass). The lot is held for 100% inspection and repair of the major class, then re-audited with a fresh sample before any shipment decision.
3. DHU = 45 ÷ 600 × 100 = 7.5. DHU is an in-line signal: it tracks the defect rate at checking points so deterioration is caught and diagnosed early; it does not itself pass or fail a shipment.

4. No single document does; the connection is the shared data set — above all the operation bulletin/SMV and the order quantity — used unchanged by all three. (Credit any answer that identifies the operation bulletin and order sheet as the common sources.)
